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characteristic of equilibrium involving physical process

a) Equilibrium is possible only in a closed system at a given temperature

b) The opposing processes occur at the same rate and there is a dynamic but stable condition

c) All the physical processes stop at equilibrium

d) All measurable properties of the system remains constant

20. For the formation of Two moles of $SO_3(g)$ from SO_2 and O_2, the equilibrium constant is K_1. The equilibrium constant for the dissociation of one mole of SO_3 into SO_2 and O_2 is

a) $1/K_1$ b) K_1^2

c) $\left(\frac{1}{K_1}\right)^{1/2}$ d) $\frac{K_1}{2}$

21. Match the equilibria with the corresponding conditions,

i) Liquid ⇌ Vapour

ii) Solid ⇌ Liquid

iii) Solid ⇌ Vapour

iv) Solute (s) ⇌ Solute (Solution)

1) melting point

2) Saturated solution

3) Boiling point

4) Sublimation point

5) Unsaturated solution

	(i)	(ii)	(iii)	(iv)
(a)	1	2	3	4
(b)	3	1	4	2
(c)	2	1	3	4
(d)	3	2	4	5

22. Consider the following reversible reaction at equilibrium, A + B ⇌ C, If the concentration of the reactants A and B are doubled, then the equilibrium constant will

a) be doubled b) become one fourth

c) be halved d) remain the same

23. $[Co(H_2O)_6]^{2+}$ (aq) (pink) + $4Cl^-$ (aq) ⇌ $[CoCl_4]^{2-}$ (aq) (blue)+ 6 H_2O (*l*)

In the above reaction at equilibrium, the reaction mixture is blue in colour at room temperature. On cooling this mixture, it becomes pink in colour. On the basis of this information, which one of the following is true ?

a) $\Delta H > 0$ for the forward reaction

b) $\Delta H = 0$ for the reverse reaction

c) $\Delta H < 0$ for the forward reaction

d) Sign of the ΔH cannot be predicted based on this information.

24. The equilibrium constants of the following reactions are :

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1

$N_2 + O_2 \rightleftharpoons 2NO$; K_2

$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O$; K_3

The equilibrium constant (K) for the reaction ;

$2NH_3 + \frac{5}{2}O_2 \overset{K}{\rightleftharpoons} 2NO + 3H_2O$, will be